

Exp No : 10 Troubleshooting Fabric Applications

Aim

To identify, analyze, and resolve common issues encountered in Hyperledger Fabric applications by troubleshooting network connectivity, chaincode deployment, channel configuration, certificate authentication, and Docker container errors.

Software Requirements

- ☐ Kali Linux / Ubuntu
- ☐ Docker
- ☐ Docker Compose
- ☐ Node.js (Version 18 or later)
- ☐ npm
- ☐ Git
- ☐ Hyperledger Fabric Samples
- ☐ Hyperledger Fabric Binaries
- ☐ Hyperledger Fabric Docker Images
- ☐ Visual Studio Code

Hardware Requirements

- ☐ Processor: Intel Core i3 or above
- ☐ RAM: Minimum 8 GB
- ☐ Storage: Minimum 20 GB Free Disk Space
- ☐ Internet Connection

Theory

Troubleshooting is an essential part of administering Hyperledger Fabric networks. Errors may occur due to incorrect configuration, network failures, certificate issues, chaincode deployment failures, or Docker container problems.

Common troubleshooting techniques include:

- ☐ Checking Docker container status
- ☐ Viewing peer and orderer logs
- ☐ Verifying channel membership
- ☐ Checking chaincode installation
- ☐ Validating MSP and certificates
- ☐ Restarting Fabric services
- ☐ Cleaning Docker resources

Proper troubleshooting minimizes downtime and ensures reliable blockchain application deployment.

Procedure

Step 1: Navigate to the Fabric Test Network
`cd ~/fabric-dev/fabric-samples/test-network`

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca
```

Expected Output

Creating channel 'mychannel'

Starting peer0.org1

Starting peer0.org2

Starting orderer.example.com

Fabric Network Started Successfully

Step 3: Verify Docker Containers

Check whether all required containers are running.

```
docker ps
```

Expected Output

peer0.org1.example.com

peer0.org2.example.com

orderer.example.com

ca_org1

ca_org2

Step 4: Check Peer Logs

Display the logs of Peer Organization 1.

```
docker logs peer0.org1.example.com
```

Expected Output

Peer Started Successfully

Listening on Port 7051

Joined Channel Successfully

Step 5: Check Orderer Logs

```
docker logs orderer.example.com
```

Expected Output
Orderer Started Successfully

Receiving Transactions

Creating New Blocks

Step 6: Verify Channel Membership
peer channel list

Expected Output
Channels peers has joined:

mychannel

Step 7: Verify Installed Chaincode
peer lifecycle chaincode queryinstalled

Expected Output
Package ID:

basic_1.0

Version:1.0

Step 8: Troubleshoot Chaincode Deployment
Redeploy the chaincode if deployment has failed.
./network.sh deployCC \
-ccl javascript \
-ccn basic \
-ccp ../asset-transfer-basic/chaincode-javascript

Expected Output
Packaging Chaincode

Installing Chaincode

Approving Chaincode

Committing Chaincode

Chaincode deployed successfully

Step 9: Verify Chaincode Execution

Query the ledger to ensure that the chaincode is functioning correctly.

```
peer chaincode query \
```

```
-C mychannel \
```

```
-n basic \
```

```
-c '{"Args":["GetAllAssets"]}'
```

Expected Output

Asset records displayed successfully.

Step 10: Check Docker Resource Usage

Monitor resource utilization.

```
docker stats
```

Expected Output

```
CONTAINER CPU % MEM USAGE
```

```
peer0.org1 3%
```

```
peer0.org2 2%
```

```
orderer 1%
```

Step 11: Restart the Fabric Network

Stop and restart the network.

```
./network.sh down
```

```
./network.sh up createChannel -ca
```

Expected Output

Network Restarted Successfully

Step 12: Clean Docker Resources (Optional)

Remove unused Docker resources.

```
docker system prune -f
```

Expected Output

Deleted Containers

Deleted Networks

Deleted Images

Total reclaimed space: xxx MB

Common Errors and Solutions

Error Message Possible Cause Solution

Error Message Possible Cause Solution

Docker daemon not running Docker service stopped `sudo systemctl start docker`

Peer connection failed Peer container stopped Restart Fabric network

Chaincode deployment

failed

Incorrect chaincode path Verify `-ccp` directory

Channel not found Channel not created

Run `./network.sh`

`createChannel`

Access denied Invalid MSP or certificates Regenerate certificates

Chaincode not installed Installation failed Redeploy chaincode

TLS handshake failed Incorrect TLS certificate Verify TLS CA files

Port already in use

Another service using the

port

Stop conflicting process

Troubleshooting Workflow

Start Fabric Network

|



Verify Docker Containers

|



Check Peer Logs

|



Check Orderer Logs

|

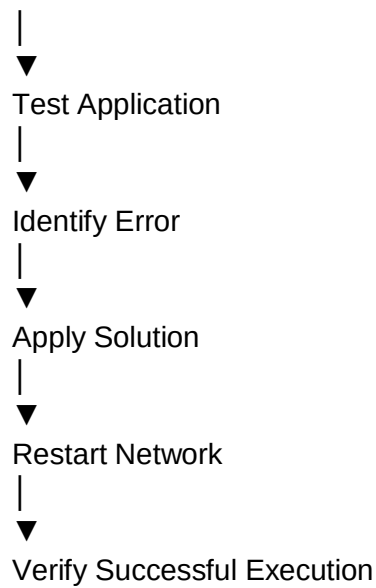


Verify Channel

|



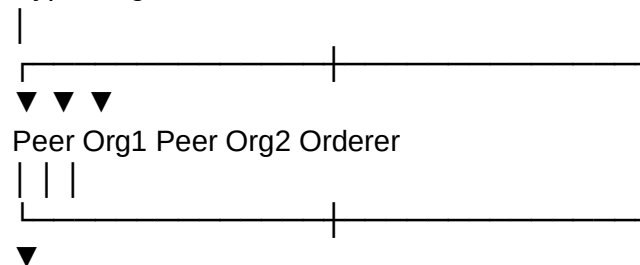
Verify Chaincode



Architecture Diagram

Client Application

Hyperledger Fabric



Blockchain Ledger

Docker Containers

Logs & Monitoring

Sample Output

Fabric Network Started Successfully

Peer Verified Successfully

Orderer Verified Successfully

Channel Verified Successfully

Chaincode Installed Successfully

Application Executed Successfully

No Errors Detected

Fabric Network Running Normally

```
(tharun@kali) ~/fabric-dev/fabric-samples/test-network
peer channel list
peer lifecycle chaincode querycommitted -C mychannel
peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
docker ps
docker stats --no-stream
docker logs --tail 50 peer0.org1.example.com
docker logs --tail 50 orderer.example.com
2026-08-20 20:18:26.073 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Channels peers has joined:
mychannel
Committed chaincode definitions on channel 'mychannel':
Name: basic, Version: 1.0, Sequence: 2, Endorsement Plugin: escc, Validation Plugin: vscc
[]
CONTAINER ID   IMAGE                                     STATUS      PORTS
NO NAMES
5759a4e1356b   dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbee664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc1497615f5e33a8bfa908e0fd3555f864cfa1398b52d341cb   "dock
er-entrypoint.s..." 7 minutes ago   Up 7 minutes
dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbee664c87b24c035bee15cbcc7f59b629   "dock
bd7407c7704b   dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbee664c87b24c035bee15cbcc7f59b629   "orde
er-entrypoint.s..." 7 minutes ago   Up 7 minutes
dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbee664c87b24c035bee15cbcc7f59b629   "sh -
e6b53923091b   hyperledger/fabric-orderer:latest      Up 13 minutes   0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:19443->19443/tcp, [::]:19443->19443/tcp
orderer.example.com
465db8b154d2   hyperledger/fabric-peer:latest         Up 13 minutes   0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
peer0.org1.example.com
b690a81f1b16   hyperledger/fabric-peer:latest         Up 13 minutes   0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
peer0.org2.example.com
bd0336956053   hyperledger/fabric-ca:latest           Up 13 minutes   0.0.0.0:7054->7054/tcp, [::]:7054->7054/tcp, 0.0.0.0:17054->17054/tcp, [::]:17054->17054/tcp
c 'fabric-ca-se..." 13 minutes ago   Up 13 minutes
ca.org1
44cef34edfba   hyperledger/fabric-ca:latest           Up 13 minutes   0.0.0.0:9054->9054/tcp, [::]:9054->9054/tcp, 7054/tcp, 0.0.0.0:19054->19054/tcp, [::]:19054->19054/tcp
c 'fabric-ca-se..." 13 minutes ago   Up 13 minutes
ca.orderer
b0a20f3dbb85   hyperledger/fabric-ca:latest           Up 13 minutes   0.0.0.0:8054->8054/tcp, [::]:8054->8054/tcp, 7054/tcp, 0.0.0.0:18054->18054/tcp, [::]:18054->18054/tcp
c 'fabric-ca-se..." 13 minutes ago   Up 13 minutes
ca.org2
CONTAINER ID   NAME                                     CPU %      MEM USAGE / LIMIT   MEM %      NET I/O      BLOCK I/O
PID
5759a4e1356b   dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbee664c87b24c035bee15cbcc7f59b629   0.00%      49.57MiB / 2GiB      2.42%      337kB / 35.2kB    0B / 4.1kB
24
bd7407c7704b   dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5fbee664c87b24c035bee15cbcc7f59b629   0.00%      51.97MiB / 2GiB      2.54%      347kB / 38.8kB    0B / 4.1kB
24
e6b53923091b   orderer.example.com                  0.09%      25.11MiB / 15.33GiB   0.16%      138kB / 312kB     0B / 332kB
15
```

```

(tharun@kali)~/fabric-dev/fabric-samples/test-network
$ ./network.sh deployCC \
  -ccl javascript \
  -con basic \
  -ccp ../asset-transfer-basic/chaincode-javascript
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAMES: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-javascript
- CC_SRC_LANGUAGE: javascript
- CC_VERSION: 1.0
+ [ -f false = true ]
+ peer lifecycle chaincode package basic.tar.gz --path ../asset-transfer-basic/chaincode-javascript --lang node --label basic_1.0
+ res=0
Chaincode is packaged
Installing chaincode on peer0.org1...
Using organization 1
+ peer lifecycle chaincode queryinstalled --output json
+ jq -r 'try (.installed_chaincodes[0].package_id)'
+ grep '^basic_1.0:1d4d445573112813889f87c307bc2b5f6e64c87b24c035bee15cbcc7f59b629$'
+ test 0 -ne 0
basic_1.0:1d4d445573112813889f87c307bc2b5f6e64c87b24c035bee15cbcc7f59b629
scripts/envVar.sh: line 86: [: too many arguments
Chaincode is installed on peer0.org1
Install chaincode on peer0.org2...
Using organization 2
+ peer lifecycle chaincode queryinstalled --output json
+ jq -r 'try (.installed_chaincodes[0].package_id)'
+ grep '^basic_1.0:1d4d445573112813889f87c307bc2b5f6e64c87b24c035bee15cbcc7f59b629$'
+ test 0 -ne 0
basic_1.0:1d4d445573112813889f87c307bc2b5f6e64c87b24c035bee15cbcc7f59b629
scripts/envVar.sh: line 86: [: too many arguments
Chaincode is installed on peer0.org2
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ sed -n '/Version:/{s;*Sequence: //; s/, Endorsement Plugin:.*$//; p;}'
+ COMMITTED_CC_SEQUENCE=1
+ res=0
+ peer lifecycle chaincode queryapproved --channelID mychannel --name basic

(tharun@kali)~/fabric-dev/fabric-samples/test-network
$ cd ../fabric-dev/fabric-samples/test-network

(tharun@kali)~/fabric-dev/fabric-samples/test-network
$ export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin
export FABRIC_CFG_PATH=$HOME/fabric-dev/fabric-samples/config

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID=Org1MSP
export CORE_PEER_TLS_ROOTCERT_FILE=$PWD/organizations/peerOrganizations/org1.example.com/peers/peer0.org1.example.com/tls/ca.crt
export CORE_PEER_MSPCONFIGPATH=$PWD/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051

(tharun@kali)~/fabric-dev/fabric-samples/test-network
$ echo $CORE_PEER_LOCALMSPID
echo $CORE_PEER_MSPCONFIGPATH
echo $CORE_PEER_ADDRESS
Org1MSP
/home/tharun/fabric-dev/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/users/Admin@org1.example.com/msp
localhost:7051

(tharun@kali)~/fabric-dev/fabric-samples/test-network
$ docker ps
CONTAINER ID   IMAGE                                CREATED      STATUS      PORTS
ID            NA
MES
5759a4e1356b   dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f6e64c87b24c035bee15cbcc7f59b629-c68f8e2a0ff0cc1497615f5e33a8bfa908e0fd3555f864cf1398b52d341cb   "dock
de
er-entrypoint.s"   3 minutes ago   Up 3 minutes
v-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f6e64c87b24c035bee15cbcc7f59b629
bd7407c7704b   dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f6e64c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433ac0e74bfaade1cf0b97f5c1163835f604aa885d60dc   "dock
de
er-entrypoint.s"   3 minutes ago   Up 3 minutes
v-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f6e64c87b24c035bee15cbcc7f59b629
e0b53923091b   hyperledger/fabric-orderer:latest
rer"           9 minutes ago   Up 9 minutes   0.0.0.0:7050→7050/tcp, [::]:7050→7050/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→9443/tcp, [::]:9443→9443/tcp   "orde
or
derer.example.com
465db8b154d2   hyperledger/fabric-peer:latest
node start"    9 minutes ago   Up 9 minutes   0.0.0.0:7051→7051/tcp, [::]:7051→7051/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp   "peer
pe
er0.org1.example.com
b090a01f1b16   hyperledger/fabric-peer:latest
node start"    9 minutes ago   Up 9 minutes   0.0.0.0:9051→9051/tcp, [::]:9051→9051/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp   "peer
pe
er0.org2.example.com
bd0336956053   hyperledger/fabric-ca:latest
c "fabric-ca-se-" 9 minutes ago   Up 9 minutes   0.0.0.0:7054→7054/tcp, [::]:7054→7054/tcp, 0.0.0.0:17054→17054/tcp, [::]:17054→17054/tcp   "sh -
ca
org1
44cef34edfba   hyperledger/fabric-ca:latest
c "fabric-ca-se-" 9 minutes ago   Up 9 minutes   0.0.0.0:9054→9054/tcp, [::]:9054→9054/tcp, 7054/tcp, 0.0.0.0:19054→19054/tcp, [::]:19054→19054/tcp   "sh -
ca
orderer
b0a20f3dbb85   hyperledger/fabric-ca:latest
c "fabric-ca-se-" 9 minutes ago   Up 9 minutes   0.0.0.0:8054→8054/tcp, [::]:8054→8054/tcp, 7054/tcp, 0.0.0.0:18054→18054/tcp, [::]:18054→18054/tcp   "sh -
ca
org2

(tharun@kali)~/fabric-dev/fabric-samples/test-network

```

Result

The Hyperledger Fabric application was successfully analyzed and common operational issues were identified and resolved using Docker commands, Fabric CLI tools, and log analysis. The experiment demonstrated effective troubleshooting techniques for network connectivity, chaincode deployment, channel configuration, certificate management, and resource monitoring.